

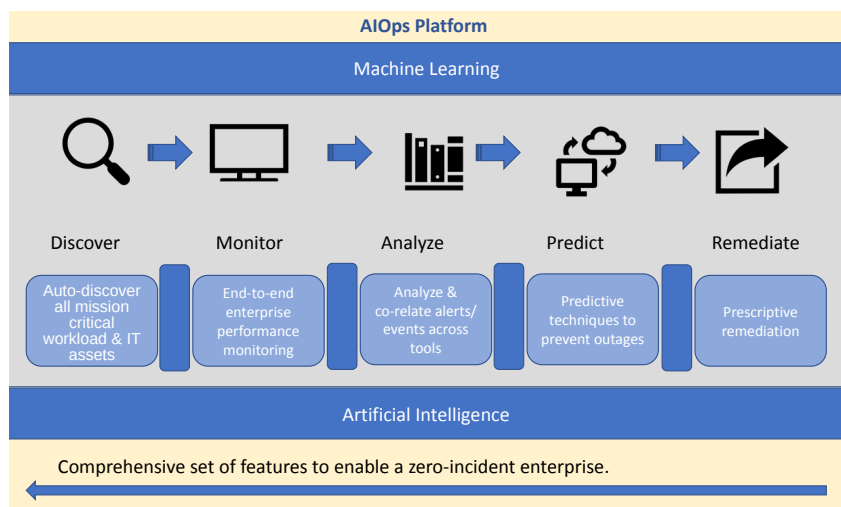


Figure1: DevOps with AIOps platform

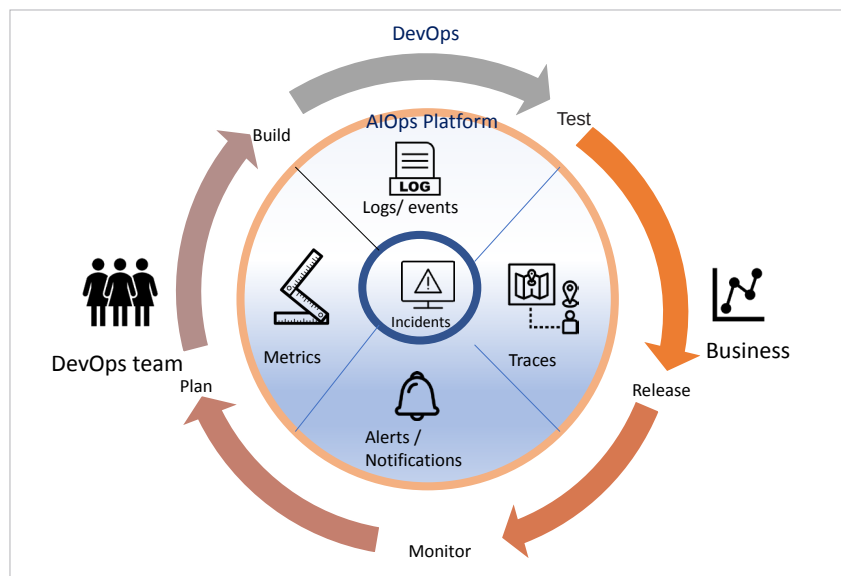


Figure 2: AIOps platform

With IT at the heart of digital transformation efforts, AIOps lets organisations operate at the speed they need to.

Union of DevOps with AIOps

DevOps is the set of practices that combines development (Dev) and IT operations (Ops) with the union of people, processes and technology to continually provide value to customers. DevOps helps the team to be highly performing, building better products faster for customer satisfaction.

DevOps gives the ownership, support and success of services to the developers that write the code. Small DevOps engineers' teams with primary site reliability engineering (SRE) give insights to correct and strengthen the reliability and scalability of services. The challenge is to get the right people across geographically dispersed teams to understand and resolve the issues of monitoring vast amounts of data.

AIOps addresses the operational challenges and covers every aspect of an organisation's service strategy. It helps in

releasing people from operations to focus on mission-critical tasks, empowering them to build improved services for better customer experiences.

AIOps in open source

Most open source AIOps projects use Python, as it is the first programming language for machine learning. Based on an organisation's thrust on operational efficiency, various AIOps and open source tools can be combined and used on AIOps platforms.

Top 5 open source AIOps tools on GitHub (based on stars)

1. SeldonIO/Seldon-core (stars: 2.2k)

This is an open source platform to deploy an organisation's machine learning models on Kubernetes at a massive scale; it has over 2 million installs.

An MLOps framework to package, deploy, monitor and manage thousands of production machine learning models, Seldon core converts your ML models (TensorFlow, PyTorch, H2O, etc) or language wrappers (Python, Java, etc) into production REST/GRPC microservices. Seldon handles scaling to thousands of production machine learning models and provides advanced machine learning capabilities out-of-the-box, including advanced metrics, request logging, explainers, outlier detectors, A/B tests, canaries, and more.

2. Logpai/Loglizer (stars: 781)

Loglizer is a machine learning based log analysis toolkit for automated anomaly detection. Logs are imperative in the development and maintenance process, as they allow developers and support engineers to monitor systems and track abnormal behaviours/errors. Loglizer provides a toolkit that implements a number of machine learning based log analysis techniques that have multiple supervised and